Hanson Wen

Computational Biology, Guinness World Record Holder

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Education

UC Berkeley — Molecular and Cell Biology + Applied Mathematics — Cumulative GPA: 3.8 | 2025–2029

- *Activities:* Project Lead at Computational Biology @ Berkeley, iGEM@Berkeley Computational Team Member (De novo protein binder design, transcriptome foundational models), ACE (Association of Chinese Entrepreneurs), AI Entrepreneurs at Berkeley Incubee SP26

Harrow International School Hong Kong — 2012–2025

- A-Levels (A* AAA): Math, Further Math, Physics, Biology; SAT: 1570

Professional Experience

Regeneron Pharmaceuticals — JEDI Research Fellowship — Summer 2026

- Developing predictive models to optimize clinical trial design and patient cohort stratification, leveraging multi-omic data and personalized biology for targeted drug efficacy

UC Berkeley Sky Computing Lab — Student Researcher — Mar 2026–Present

- rLLM hive team. Developing Agent Swarms for Biology. [Repo]

Berkeley RDI — Student Researcher — Jan 2026–Present

- Helping build AgentHLE, Humanity's Last Exam for computer use agents; benchmarking agent performance across 63 high-GDP domains from 3D modeling to accounting to education

Enso Biosciences — ML Engineer — Nov 2025–Present

- Building foundational AI models that leverage multi-modal patient input (transcriptomics, histopathology slides, EHR etc.) to help oncologists better predict treatment risks for cancer patients
- Model training/inference infrastructure, model architecture design

CytoAurora Biotechnology — Research/ML Intern — Jun–Aug 2025

- Contributed to CytoAurora's AI-driven computer vision screening pipeline for circulating tumor cell (CTC) detection from liquid biopsies; worked with nano-biological chip imaging data and assisted in training and evaluating cell classification models for automated histopathology/cytopathology analysis

Research & Projects

- **Personal Genome Analysis:** Processed my own genome (131GB WGS data), achieving 88.63% mapping rate; identified 4.99M variants including 471 high-impact mutations across 355 genes; characterized East Asian ancestry (2.08M population-consistent variants); analyzed pharmacogenomic markers (CYP2D6, CYP2C19, CYP2C9) affecting drug metabolism; deployed pipeline on Google Cloud Platform requiring 128GB RAM/1.5TB storage [Repo]
- **Enso Atlas:** Google MedGemma Impact Challenge Top 10 (Honorable Mention) out of 878 submissions; fully local on-premise pathology AI system for treatment-response prediction; analyzes whole-slide images using TransMIL, Path Foundation, MedSigLIP, and MedGemma for slide-based prediction, evidence visualization/retrieval, and reporting; FastAPI + Next.js + PostgreSQL
- **De Novo Protein Binder Design & Validation:** Built RFdiffusion-based pipeline for generating novel protein structures conditioned on active site motifs; validated designs via structural alignment of catalytic residues using PyMOL and binding affinity scoring with AutoDock Vina; developed as part of iGEM computational team

Publications

- **The Hallucination Dependence Index: A Cross-Condition Diagnostic for Clinical-LLM Faithfulness** (co-first author) — ICML 2026 GenBio Workshop, accepted poster: clinical-LLM faithfulness metric measuring how grounding suppresses unsupported hallucinations across frontier LLMs on TCGA-LUAD oncology cases
- **WorldFork: Trace-Auditable Forecasting Agents in Open-Ended Domains** (first author) — ICML 2026 (AIWILD): trace-auditable forecasting agent workflow that turns public event cards into branching timelines with endpoint ledgers, path mass, and report provenance; pilot evaluation on 24 masked retrospective resolved-event cards
- **Arthritis Biomarker Machine Learning** (first author) (DOI: 10.33696/Proteomics.4.015): Identified 79 DEGs across spondyloarthritis and rheumatoid arthritis from spatial transcriptomics of 6 synovial biopsies; 93.3% classification accuracy (LGBM); GO enrichment, KEGG pathway, and SHAP interpretation

Honors & Awards

- **Guinness World Record** — Fastest scheduled-flight circumnavigation (44h 33m 39s); partnered with disability rights advocate Cham-Kai Yip to promote accessible air travel; featured live on TVB — 2024
- **iGEM Gold Medal (Team Lead)** — Led 14-member synthetic biology team to Gold Medal; coordinated wet lab experiments, computational modeling, and human practices components — 2022
- **Hackathon Wins:** **HackTech 2026** — 1st Listen Labs; **CalHacks 12.0** — 1st Regeneron; **YC Bio/AI** — Winner; **MedGemma Impact Challenge** — Top 10 / Honorable Mention — 2025–2026